# PYTHON BASICS

A Short Beginner's Guide

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# 1. What is Python?

Python is a general-purpose programming language known for readable syntax. People use it to automate tasks, build websites, analyze data, and create prototypes. A program is a set of instructions that a computer executes.

When learning a language, focus first on small ideas and practise them. Read an example, predict what it will do, run it in a Python environment, and then change one part to see what happens.

> Learning tip: Understanding a short program is more valuable than memorizing a long one.

# 2. Your first program

The print() function displays information. Try this code:

```
print("Hello, BookNest!")
```

Output:

```
Hello, BookNest!
```

Text surrounded by quotation marks is called a string. The parentheses contain the value passed to print(). Change the message and run the program again.

# 3. Variables and data types

A variable is a name that refers to a value. Python uses the value to determine its type.

```
reader_name = "Asha"\nbooks_read = 3\nreading_goal_met =
False\nprint(reader_name)\nprint(books_read)
```

Here, reader_name refers to text, books_read refers to an integer, and reading_goal_met refers to a Boolean value. Use names that explain what the data represents.

## A simple calculation

```
price = 99\nquantity = 2\ntotal = price * quantity\nprint(total)
```

The output is 198. The multiplication operator * calculates the total for two items at the same price.

# 4. Conditions

Conditions let a program choose between different actions. The if statement runs its indented block when its condition is true.

```
score = 72\nif score >= 50:\n print("Passed")\nelse:\n print("Keep practising")
```

Because score is 72, the program prints Passed. Notice the colon and indentation. In Python, indentation is part of the program structure.

# 5. Loops

A loop repeats instructions. A for loop is useful when you want to do something once for each item in a sequence.

```
for chapter in range(1, 4):\n print("Chapter", chapter)
```

This prints Chapter 1, Chapter 2, and Chapter 3. The range(1, 4) sequence starts at 1 and stops before 4.

### A while loop

```
count = 1\nwhile count <= 3:\n print(count)\n count += 1
```

The variable increases on every pass, so the condition eventually becomes false. Always check that a while loop can finish.

# 6. Functions

A function groups instructions under a name so they can be reused. Define one with def.

```
def welcome(reader):\n return "Welcome to BookNest, " + reader + "!"\n\nmessage = welcome("Asha")\nprint(message)
```

The function receives a parameter called reader and returns a string. Calling welcome('Asha') produces a personalized message.

# 7. Mini project: tip calculator

This small program calculates a tip and the total bill. It uses numeric input, multiplication, and addition.

```
bill = 500\ntip_rate = 0.10\ntip = bill * tip_rate\ntotal = bill + tip\nprint('Tip:', tip)\nprint('Total:', total)
```

The output is Tip: 50.0 and Total: 550.0. Try changing bill to 850 or tip_rate to 0.15. A real version could ask the user to enter these values.

# 8. Practice questions and answers

### Try these questions

1. Which function displays text in Python?

2. What value does 5 * 4 produce?

3. How many times does range(1, 5) run in a for loop?

4. Which keyword starts a function definition?

## Answers

1. print(). 2. 20. 3. Four times, for 1, 2, 3, and 4. 4. def.

Practice method: Before running code, write down what you expect to happen. Then compare your prediction with the actual output.

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